

AI/ML Internship Task 4 Report

Breast Cancer Classification: Model Evaluation and Imbalanced Data Handling

Objective

The objective of this task was to develop a **binary classification model** using the Breast Cancer dataset and evaluate its performance using different machine learning techniques.

The main goals of this task were:

- Implement a classification algorithm using Logistic Regression.
- Understand and apply classification evaluation metrics.
- Analyze model performance using Confusion Matrix, Precision, Recall, F1 Score, ROC Curve, and AUC.
- Handle potential class imbalance using class weighting.
- Compare Logistic Regression performance with a Decision Tree Classifier.

Dataset Description

For this task, the **Breast Cancer Wisconsin dataset** available through Scikit-learn was used.

The dataset contains numerical features representing characteristics of breast cancer samples.

Dataset Information:

- **Problem Type:** Binary Classification
- **Number of Features:** 30 numerical features
- **Target Classes:**
 - 0 → Malignant
 - 1 → Benign

Before model development, the dataset was explored to understand:

- Dataset shape
- Feature information
- Target classes
- Class distribution

Data Preprocessing

Train-Test Split

The dataset was divided into training and testing sets using the train-test split technique.

Parameters used:

- Test size: 20%
- Random state: 42
- Stratified split was applied to maintain the proportion of classes in both training and testing data.

Feature Scaling

Feature scaling was performed using **StandardScaler**.

Scaling was applied because Logistic Regression performs better when input features are normalized to a similar range.

Steps performed:

- Fit scaler on training data.
- Transform training and testing features.

Classification Models Implemented

Three classification models were implemented and evaluated.

Logistic Regression

Logistic Regression was used as the baseline classification model.

The model was trained using scaled features and used to predict whether a sample belongs to the malignant or benign class.

Balanced Logistic Regression

To handle possible class imbalance, Logistic Regression was implemented with:

```
class_weight='balanced'
```

This technique automatically adjusts class weights based on the frequency of each class.

Decision Tree Classifier

A Decision Tree Classifier was implemented as a comparison model.

The objective was to compare the performance of tree-based classification with Logistic Regression.

Model Evaluation Metrics

The models were evaluated using multiple classification metrics:

Accuracy

Measures the overall percentage of correctly classified samples.

Precision

Measures how many predicted positive cases were actually positive.

Recall

Measures how many actual positive cases were correctly identified.

F1 Score

The harmonic mean of Precision and Recall.
It provides a balanced evaluation of model performance.

Confusion Matrix

A confusion matrix was generated to analyze:

- True Positives
- True Negatives
- False Positives
- False Negatives

ROC Curve and AUC

ROC Curve was plotted to visualize the trade-off between:

- True Positive Rate
- False Positive Rate

AUC score was calculated to measure the model's classification capability.

Model Performance Comparison

The performance comparison of implemented models is shown below:

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.982456	0.986111	0.986111	0.986111
Balanced Logistic Regression	0.956140	0.985507	0.944444	0.964539
Decision Tree	0.912281	0.955882	0.902778	0.928571

Results and Analysis

Based on the evaluation results:

Logistic Regression

- Achieved the highest accuracy.
- Achieved the highest F1 Score.
- Provided the best overall classification performance.

Balanced Logistic Regression

- Demonstrated the impact of class weighting.
- Improved handling of possible class imbalance.

- Maintained strong precision and recall performance.

Decision Tree Classifier

- Provided a comparison with a different classification approach.
- Achieved good performance but lower than Logistic Regression for this dataset.

Best Performing Model

Based on the evaluation metrics, especially **F1 Score and Accuracy**, the best-performing model was:

Logistic Regression

Performance:

- Accuracy: **98.24%**
- Precision: **98.61%**
- Recall: **98.61%**
- F1 Score: **98.61%**

The model demonstrated strong classification ability for identifying breast cancer classes.

Conclusion

In this task, a complete binary classification workflow was implemented using the Breast Cancer dataset.

The project included:

- Data preprocessing
- Feature scaling
- Logistic Regression implementation
- Handling class imbalance
- Model comparison
- Classification performance evaluation

Different evaluation metrics such as Accuracy, Precision, Recall, F1 Score, Confusion Matrix, ROC Curve, and AUC were used to analyze model performance.

Among the evaluated models, **Logistic Regression achieved the best results**, making it the selected final model for this classification problem.

This task enhanced practical understanding of classification algorithms, evaluation techniques, and machine learning model comparison.

Tools & Technologies Used:

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- Jupyter Notebook